

TDDE58 - Lab Description

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In this lab, you will implement a multiple-access setup with two users. The goal of this lab is to understand how two simple yet important medium access control protocols operate, evaluate their performance, and explain their advantages and limitations.

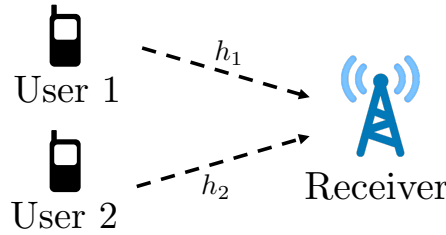


Figure 1: A wireless network with two users sharing the same channel to communicate with a receiver.

We consider a setup with two users transmitting information in the form of packets over a common wireless channel to a receiver as shown in Fig. 1. The time is assumed to be slotted. We denote by $P_{\text{tx},i}$ the transmission power of user $i, i = 1, 2$. The received power for the transmission of user i is $P_{\text{rx},i} = P_{\text{tx},i} h_i r_i^{-\alpha}$ $i = 1, 2$. h_i is the channel power gain and is a random variable representing the fading. More specifically, the channel power gains, h_1 and h_2 are independent exponential random variables with mean one. They remain constant during a slot and are independently redrawn in each new slot. In addition, r_i is the distance between user i and the receiver, and α is the path loss exponent. Here we consider $\alpha = 4$ and $r = r_1 = r_2$.

A transmission is successful if the Signal-to-Noise-Ratio (SNR) (in case only one user transmits) or the signal-to-interference-and-noise ratio (SINR) (if both users transmit at the same time) is above a certain threshold γ .

1 Medium access control protocols

We will investigate the performance of TDMA and slotted ALOHA protocols. Below we give a brief description, for a more detailed treatment please refer to the textbook.

1.1 TDMA

We consider the TDMA protocol where user 1 transmits during the odd slots, and user 2 transmits during the even slots. Note that in TDMA, there is no interference among the users, thus the transmissions are noise limited and only the SNR needs to be considered. The received SNRs for users 1 and 2 are

$$\text{SNR}_1 = \frac{P_{\text{tx},1} h_1 r^{-4}}{N_0}, \quad (1)$$

and

$$\text{SNR}_2 = \frac{P_{\text{tx},2} h_2 r^{-4}}{N_0}, \quad (2)$$

respectively. N_0 is the thermal noise power at the receiver. A packet from user i is successfully transmitted only if $SINR_i > \gamma$ in its intended timeslot.

1.2 Slotted ALOHA

In the slotted ALOHA protocol, the users transmit with a probability q at every slot. For simplicity, here we consider the case $q = 1$, which means that both users transmit and interfere with each other in every time slot. Thus, we need to calculate the received SINR for each user as given below

$$SINR_1 = \frac{P_{tx,1}h_1r^{-4}}{P_{tx,2}h_2r^{-4} + N_0}, \quad (3)$$

and

$$SINR_2 = \frac{P_{tx,2}h_2r^{-4}}{P_{tx,1}h_1r^{-4} + N_0}. \quad (4)$$

The transmission from user i in a timeslot is successful if $SINR_i > \gamma$.

2 Tasks

Using the programming language or tool of your preference, you need to calculate and plot the network throughput for the two protocols for a set of cases given below. We consider that the noise power is $N_0 = 10^{-9}$ mW. Assume that both users always have packets available for transmission and that each packet occupies exactly one time slot. Ignore acknowledgment and other control overhead. For each parameter combination, simulate at least 10^5 time slots and estimate the network throughput as the total number of successfully received packets from both users divided by the number of simulated slots. Use a fixed random seed for reproducibility and state it in your report.

1. Plot the network throughput versus the common transmission power $P = P_{tx,1} = P_{tx,2}$, using $P \in \{1, 2, 4\}$ mW. Create one figure for each threshold:

- (a) $\gamma = 0.2$,
- (b) $\gamma = 1$,
- (c) $\gamma = 3$.

Each figure should contain four curves: TDMA at $r = 200$ m, TDMA at $r = 400$ m, ALOHA at $r = 200$ m, and ALOHA at $r = 400$ m. Each curve should contain three simulated points, corresponding to the three transmission powers. Label the x-axis “Transmission power (mW)” and the y-axis “Network throughput (packets/slot)”, and include a legend.

2. Provide a detailed discussion for the three figures. When does TDMA outperform ALOHA? Explain your observations and discuss how transmission power and distance affect the network throughput.
3. Choose one of the scenarios above and plot the performance versus distance r for fixed power, explain what you observe and why you selected the range values for the distance.
4. (For groups of 2) Relax the condition $q = 1$ and consider varying it. What do you observe? Provide a detailed explanation, and consider the scenario that different users have different probabilities of accessing the channel. Assume that transmission decisions are independent across users and time slots, and independent of the channel gains. When only one user transmits, determine its reception outcome using the corresponding SNR. When both users transmit, evaluate each packet separately using its corresponding SINR. When neither user transmits, the slot is idle and contributes zero successful receptions.